Chapter 4: DC Circuits

4.1 Resistance and Ohm's law

A resistor opposes the flow of current. Ohm's law states that the potential difference across a resistor is proportional to the current through it: V = I R.

4.2 Resistors in series and parallel

Resistors in series carry the same current; their resistances add. Resistors in parallel have the same potential difference across them; the reciprocals of their resistances add.

4.3 Measuring instruments

A voltmeter is connected in parallel with the element being measured. An ideal voltmeter has infinite resistance, so it draws no current and does not disturb the circuit. An ammeter is connected in series and an ideal ammeter has zero resistance.

Worked example. A voltmeter is in series with a 9 ohm resistor and the pair is connected across a 3 ohm resistor. Because the voltmeter draws no current, the 9 ohm resistor has no voltage drop, so the voltmeter reads the potential difference across the 3 ohm resistor.